

llama2_7b (meta-llama/Llama-2-7b-hf)

Table 1 — CCS accuracy (mean $\pm$ sd over seeds)

dataset	answer (%)	n_answer	period (%)	n_period
comparisons	100 $\pm$ 00	30	93 $\pm$ 01	30
sp_en_trans	100 $\pm$ 00	30	95 $\pm$ 12	30
cities	100 $\pm$ 00	30	100 $\pm$ 00	30
amazon	96 $\pm$ 00	30	93 $\pm$ 00	30
imdb	94 $\pm$ 00	30	92 $\pm$ 01	30
ent_bank	84 $\pm$ 09	30	81 $\pm$ 19	30
snli	64 $\pm$ 08	30	82 $\pm$ 07	30
copa	58 $\pm$ 06	30	48 $\pm$ 02	30
rte	54 $\pm$ 04	30	52 $\pm$ 03	30

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Table 2 — loss-term ablations and alterations (answer token)

dataset	CCS	L_conf	L_cons	L_cons+a1	L_cons+a2	L_cons+a1+a2	CCS+a1+a2
comparisons	100 ± 00	99 ± 01	65 ± 12	65 ± 13	67 ± 11	60 ± 07	100 ± 00
sp_en_trans	100 ± 00	96 ± 08	70 ± 13	70 ± 13	66 ± 10	68 ± 10	100 ± 00
cities	100 ± 00	97 ± 09	69 ± 11	73 ± 13	69 ± 14	63 ± 10	98 ± 03
amazon	96 ± 00	95 ± 00	67 ± 12	70 ± 12	74 ± 13	66 ± 10	96 ± 00
imdb	94 ± 00	88 ± 06	61 ± 08	61 ± 07	63 ± 10	60 ± 07	94 ± 00

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Table 3 — tcPCA vs the spread of CCS solutions (period token)

dataset	CCS min	CCS med	CCS max	CRC-TPC	tcPCA	GCC
comparisons	93	93	94	52	94	96
sp_en_trans	56	99	99	99	99	99
cities	100	100	100	51	99	100
amazon	92	93	93	51	93	93
imdb	90	92	93	91	92	93
ent_bank	50	93	96	89	90	89
snli	68	84	92	75	81	84
copa	44	48	55	50	51	52
rte	47	52	59	52	52	52

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Table 5 — accuracy of individual principal components

dataset	token	PC1	PC2	PC3	PC4	PC5
comparisons	answer	100	51	41	52	56
sp_en_trans	answer	100	52	44	60	55
cities	answer	100	60	53	51	58
amazon	answer	96	54	50	49	54
imdb	answer	93	56	54	46	54
ent_bank	answer	85	48	59	58	56
snli	answer	63	61	76	64	55
copa	answer	51	70	44	57	53
rte	answer	52	65	50	61	52
comparisons	period	52	55	48	93	56
sp_en_trans	period	99	54	46	51	42
cities	period	51	95	65	61	57
amazon	period	51	92	50	49	54
imdb	period	91	60	51	48	52
ent_bank	period	89	52	55	64	56
snli	period	75	75	54	64	54
copa	period	50	49	47	58	53
rte	period	52	61	50	60	53

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Sign instability — how often the learned orientation came out backwards

dataset	answer flipped (%)	period flipped (%)
comparisons	39	44
sp_en_trans	54	51
cities	50	44
amazon	36	46
imdb	43	52
ent_bank	46	52
snli	44	50
copa	52	54
rte	44	51